Deep Learning for Economics and Finance

Surrogate-based simulated method of moments in the Brock–Mirman model

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Solution notebooks:

`03_Structural_Estimation_BM.ipynb` (single ϱ)

`03b_Structural_Estimation_BM_Joint.ipynb` (joint β, ϱ)

Materials:

`https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models`

The Model: Brock–Mirman with Stochastic TFP

Stochastic growth model:

$$\max_{\{C_t\}} \sum_{t=0}^{\infty} \beta^t \mathbb{E}[\ln C_t], \qquad K_{t+1} + C_t = Y_t + (1-\delta)K_t$$

$$Y_t = z_t K_t^{\alpha}, \qquad \log z_{t+1} = \varrho \log z_t + \sigma\, \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0,1).$$

Calibrated parameters

α	δ	σ	β (single-ϱ exercise)
0.36	0.10	0.04	0.96

Pedagogical sequence:

- **Exercise 1**: estimate $\varrho \in [0.50, 0.99]$ from autocorrelation moments. Sharp 1D identification.
- **Exercise 2**: estimate (β, ϱ) *jointly*. Visualize identification ridge; collapse it with extra moments.

Why ϱ, not β?

- β is famously **poorly identified** from macro moments alone – it shifts the level of the savings rate but every level moment depends on the same scalar.
- In practice, β is calibrated to a target interest rate ($1/\beta \approx 1 + r$), *not* estimated from data.
- ϱ has a **clean direct mapping** to autocorrelation moments: $\mathrm{Corr}(\log Y_t, \log Y_{t-1})$, $\mathrm{Corr}(\Delta \log C_t, \Delta \log C_{t-1})$, and the simulated volatility moment move with persistence.
- This is the textbook example of a **just-identified** parameter, which is exactly what we want for teaching the SMM workflow.

The single-parameter exercise (notebook 03) recovers $\hat{\varrho}$ accurately under CRN. The joint exercise (notebook 03b) shows what happens when an additional, partially-identified parameter is freed.

Task 1: Train Surrogate with ϱ as Pseudo-State

Goal

Train $\mathcal{N}(z_t, K_t, \varrho) \to s_t \in (0,1)$, with $K_{t+1} = (1-\delta)K_t + Y_t \cdot s_t$.

1. 3-input MLP $(z, K, \varrho) \to s$ with sigmoid output.
2. Euler residual loss with Gauss–Hermite quadrature; ϱ enters the *transition* as well as the policy.

$$\mathcal{L} = \frac{1}{N} \sum_i \left(1 - \beta C_t^{(i)} \sum_j w_j \frac{1 - \delta + \alpha z_{t+1,j}^{(i)} K_{t+1}^{(i)\,\alpha-1}}{C_{t+1,j}^{(i)}} \right)^2$$

3. Two-phase training: (i) uniform sampling on (z, K, ϱ) box, (ii) simulation-based refinement.
4. Validate by plotting $s(z = 1, K, \varrho)$ for several ϱ.

Task 2: Simulate and Compute Identifying Moments

Goal

Generate synthetic "ex-post" data at $\varrho_{\text{true}} = 0.90$ and compute three identifying moments.

Moment	What it captures
$m_1 = \mathrm{Std}(\Delta \log C_t)$	Growth-rate volatility; check numerically
$m_2 = \mathrm{Corr}(\Delta \log C_t, \Delta \log C_{t-1})$	Persistence in consumption growth
$m_3 = \mathrm{Corr}(\log Y_t, \log Y_{t-1})$	Persistence in output (very direct)

Note: $\mathrm{Var}(\log z) = \sigma^2/(1-\varrho^2)$, but $\mathrm{Var}(\Delta \log z) = 2\sigma^2/(1+\varrho)$.

CRN: same burn-in, horizon, initial state, seed for every candidate ϱ.

Task 3: Estimate ϱ via SMM

SMM objective (single-parameter case, $\theta = \varrho$)

$$\hat{\varrho} = \arg\min_{\varrho} \, [m(\varrho) - \hat{m}^{\text{data}}]^\top W \, [m(\varrho) - \hat{m}^{\text{data}}], \quad W = I.$$

1. Define one routine $\varrho \mapsto (C_t, I_t, Y_t) \mapsto m(\varrho)$ that reuses the trained surrogate.
2. Coarse grid pass on $\varrho \in [0.50, 0.99]$ to localize the minimum.
3. Refine with `scipy.optimize.minimize_scalar` (bounded Brent's method).
4. **Verify the solution is interior** ($\hat{\varrho}$ strictly inside the bounds), required for asymptotic theory.

Hint

Re-seed the simulator every evaluation. Without CRN the criterion is dominated by simulation noise.

Stacking a GP over the Moment Map

Two-layer surrogate for high-throughput SMM:

- **Layer 1.** Pseudo-state DEQN $\mathcal{N}_\rho(z, K, \theta) \mapsto$ savings rate
- **Layer 2.** One Gaussian process per moment, $\widehat{m}_j(\theta) \sim \mathrm{GP}(0, k_j)$, fit on a small design $\{(\theta^{(i)}, m(\theta^{(i)}))\}_{i=1}^n$

Cost of one design point. One T-step forward simulation under $\mathcal{N}_\rho$ + a moment computation $\Rightarrow$ expensive but parallel.

Cost of one objective evaluation after training. One GP posterior call per moment, no simulation. Bootstrap/SBI workflows then run on the GP.

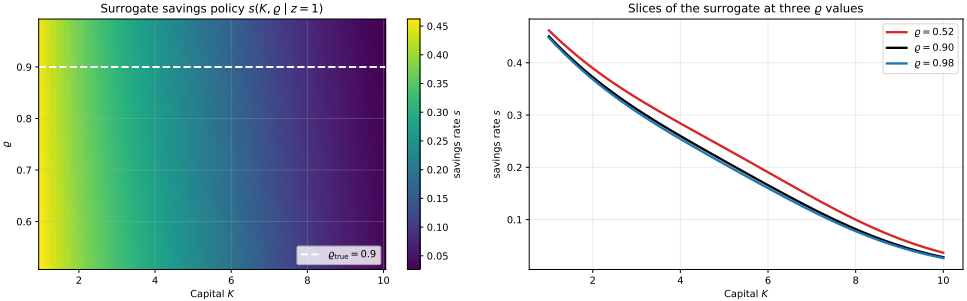
Diagnostic. Cholesky-trick LOO on each moment GP (free, see deck 07). Notebook 03: 8 grid + 6 BAL points, fresh holdout for sanity.

Same expensive-oracle / supervised-learning logic as GP-VFI in deck 07.

Task 4: Diagnose the Fit (Single-Parameter)

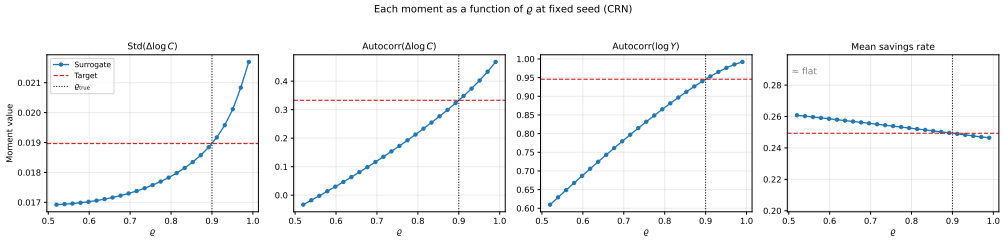
- How close is $\hat{\varrho}$ to $\varrho_{\text{true}} = 0.90$?
- Plot the SMM objective profile and mark both ϱ_{true} and $\hat{\varrho}$.
- Compare matched moments $m(\hat{\varrho})$ versus targets $\hat{m}^{\text{data}}$.
- Plot $s(K, \varrho_{\text{true}})$ vs. $s(K, \hat{\varrho})$ – the curves should overlay almost exactly.
- Trace each candidate moment as a function of ϱ on the trained surrogate – this is the identification picture.

Expected Result: Surrogate Slice $s(K, \varrho \mid z = 1)$



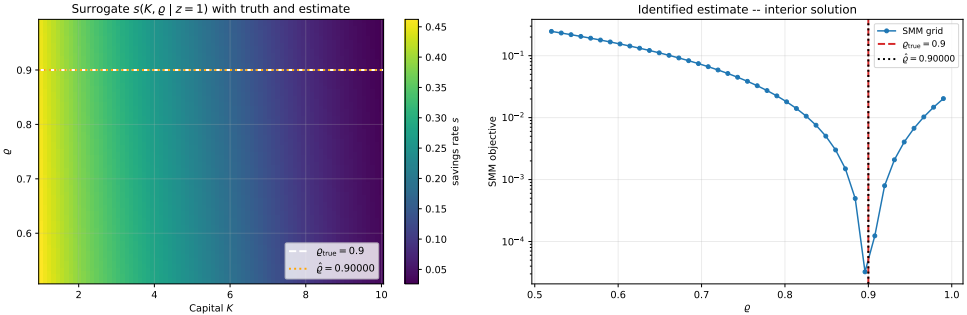
The 3-input network is a smooth function over the entire (K, ϱ) rectangle. The right panel shows three slices; the left panel shows the full surrogate compressed into one heatmap.

Expected Result: Identifying Variation



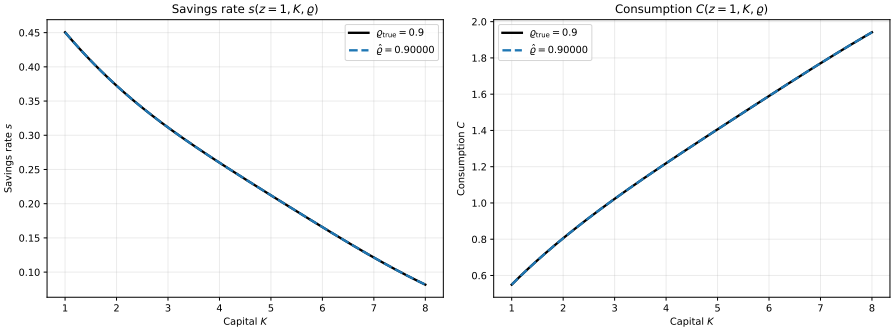
At fixed CRN, each candidate moment is monotone (or near-monotone) in ϱ *except* the mean savings rate (rightmost panel), which is essentially flat – correctly diagnosed as uninformative for ϱ identification.

Expected Result: SMM Objective + Estimate



Left: surrogate slice with truth (white dashed) and $\hat{\varrho}$ (orange dotted). Right: log-SMM-objective profile with both. The estimate sits strictly inside the parameter rectangle (**interior** solution).

Expected Result: Policy Match



The estimated policy at $\hat{\varrho}$ overlaps the true policy at $\varrho_{\text{true}} = 0.90$ to plotting accuracy.
Single-parameter SMM with sharp identification recovers the truth accurately under CRN.

Bridge: From Just-Identified to Joint

The single-parameter exercise is the cleanest version of the SMM workflow. But two real lessons remain:

- How does the surrogate scale to *multiple* parameters? Same idea: add them as pseudo-states.
- What does the SMM criterion look like when one parameter is **partially identified**?

Notebook 03b re-trains the surrogate on the rectangle $\beta \in [0.92, 0.99] \times \varrho \in [0.50, 0.99]$ and estimates both jointly.

Two moment subsets:

- **Just-identified** (2 moments): $\{$ mean savings, $\mathrm{Corr}(\Delta \log C)\}$.
- **Over-identified** (4 moments): adds $\mathrm{Std}(\Delta \log C)$ and $\mathrm{Corr}(\log Y)$.

Joint Task: Estimate (β, ϱ)

Goal

$\hat{\theta} = \arg\min_{\theta=(\beta,\varrho)} [m(\theta) - \hat{m}^{\text{data}}]^\top W [m(\theta) - \hat{m}^{\text{data}}]$ on $\beta \in [0.92, 0.99] \times \varrho \in [0.50, 0.99]$.

1. 4-input MLP $(z, K, \beta, \varrho) \to s$. Train once on the whole rectangle.
2. Forge synthetic data at $(\beta_{\text{true}}, \varrho_{\text{true}}) = (0.96, 0.90)$.
3. Compute the SMM criterion on a 25×25 grid in (β, ϱ) – visualize both moment subsets.
4. Refine with `scipy.optimize.minimize` (bounded Nelder–Mead from the grid argmin).
5. Verify the solution is interior; report distance to truth on each axis.

BoTorch UCB Acquisition for Joint Estimation

Why active learning here. On the 2D rectangle, most of the area sits far from the SMM minimiser – pure exploration is wasteful. We want a design that hits both the basin of attraction **and** regions where the moment GP is uncertain.

Acquisition function (notebook 03b):

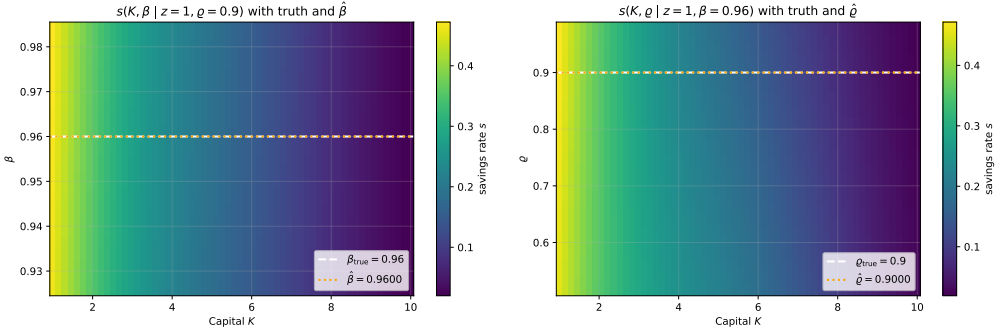
$$a(\theta) \;=\; \left[\,0.25 + \widetilde{\mathrm{UCB}}_g(\theta)\,\right] \cdot \left\| \boldsymbol{\sigma}_m(\theta)/\widehat{\mathrm{sd}}(m) \right\|_2, \quad g(\theta) = -\log_{10}(Q(\theta) + \epsilon).$$

- First factor: **exploit** – favor θ with small SMM criterion
- Second factor: **explore** – favor θ where the moment GP is uncertain
- Constant 0.25: keep exploration alive even where scaled UCB$=0$

Three-way comparison. Pilot grid / naive LHS / BoTorch BAL at the same budget: compare LOO error, criterion error, and recovered estimate.

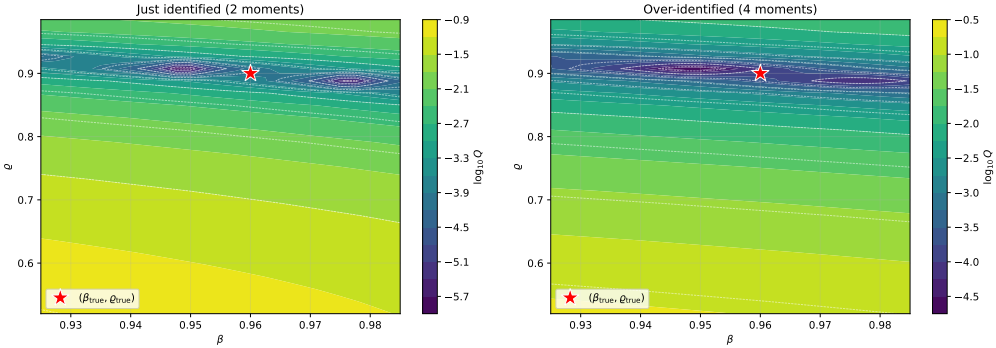
Implementation: BoTorch `SingleTaskGP` + `UpperConfidenceBound`.

Expected Result: Surrogate Slices on (β, ϱ)



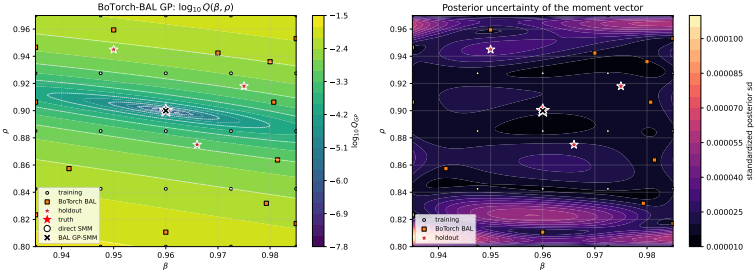
Two slices of the same 4-input network. White dashed = truth, orange dotted = over-identified estimate. The estimate sits strictly **inside** the rectangle on both axes simultaneously.

Expected Result: 2D SMM Criterion



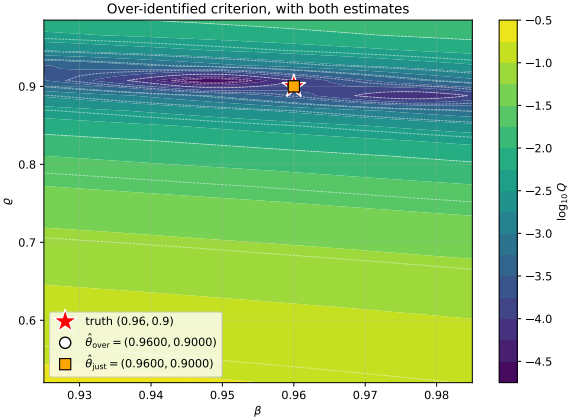
Left: the just-identified criterion shows a **ridge** along β – exactly the partial-identification problem that motivates calibrating β in practice. Right: adding two persistence moments collapses the ridge to a localized minimum.

GP Surrogate of the SMM Criterion



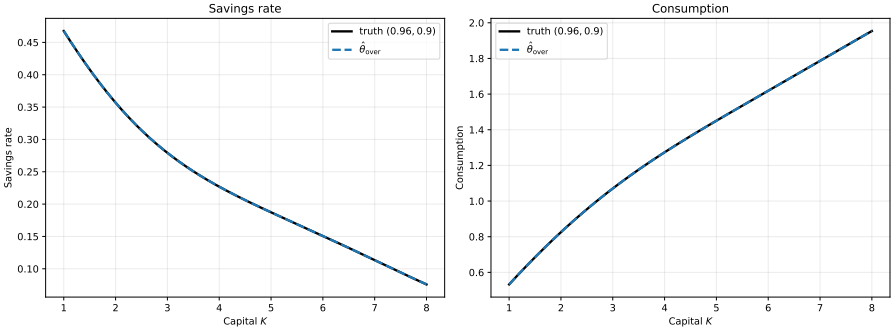
- Left: GP posterior mean of $\log_{10} Q(\beta, \varrho)$; right: standardized moment uncertainty.
- White dots = training design; orange squares = active additions; red stars = holdouts/truth.
- White circle = direct SMM; black cross = GP-SMM. Later objective calls need only one GP posterior evaluation.

Expected Result: Estimates on Top of Criterion



In this synthetic CRN run, both estimators recover (β, ϱ) close to the truth. The takeaway is the criterion's *shape*: the just-identified ridge would translate into wide confidence bands on β in real-data applications, while the over-identified criterion has localized curvature on both axes.

Joint Policy Match



Savings rate and consumption at the over-identified estimate overlay the truth almost exactly. The 4-input surrogate, trained *once*, supports millions of cheap moment evaluations during the optimizer's line-search.

Wrap-Up

What the two notebooks teach together:

- A neural surrogate with parameters as pseudo-states **decouples** the cost of solving the structural model from the cost of moment-matching. Train once, use it inside any optimizer.
- For a well-identified parameter (ϱ), SMM under CRN gives a sharp 1D identification picture.
- For **joint** estimation, the contour plot makes the partial-identification ridge visible. Adding moments collapses it.
- The **interior-solution** check is part of the standard pipeline, needed for the asymptotic distribution theory of the SMM estimator.

Formal identification machinery (moment Jacobian $M = \partial m/\partial\theta$, rank condition, J-statistic, weak-identification diagnostics): see lecture script Sec. 10.4 (Practical Considerations).

Production pipeline: Chen, Didisheim & Scheidegger (2026, JFE), exactly this surrogate-and-SMM logic at scale, applied to finance.